

HW4

Steps:

1. Randomly generate a large number of programs (initial population).
2. Evaluate (execute) each program on the test data.
3. Combine the best programs together using selection, crossover, and mutation.
4. Repeat the process to produce subsequent generations.

Input Domain: $[-5, +5]$ and the goal is to minimize the output (error).

- Internal Nodes: $\{+, -, /, \sin, \cos, *, \text{positive}, \text{ifelse}\}$
- Terminal Nodes: $\{x1, x2, 0.0, 1.0\}$

To generate the initial population, we use the Ramped half-and-half method with depths ranging from 2 to 4, which ensures high diversity.

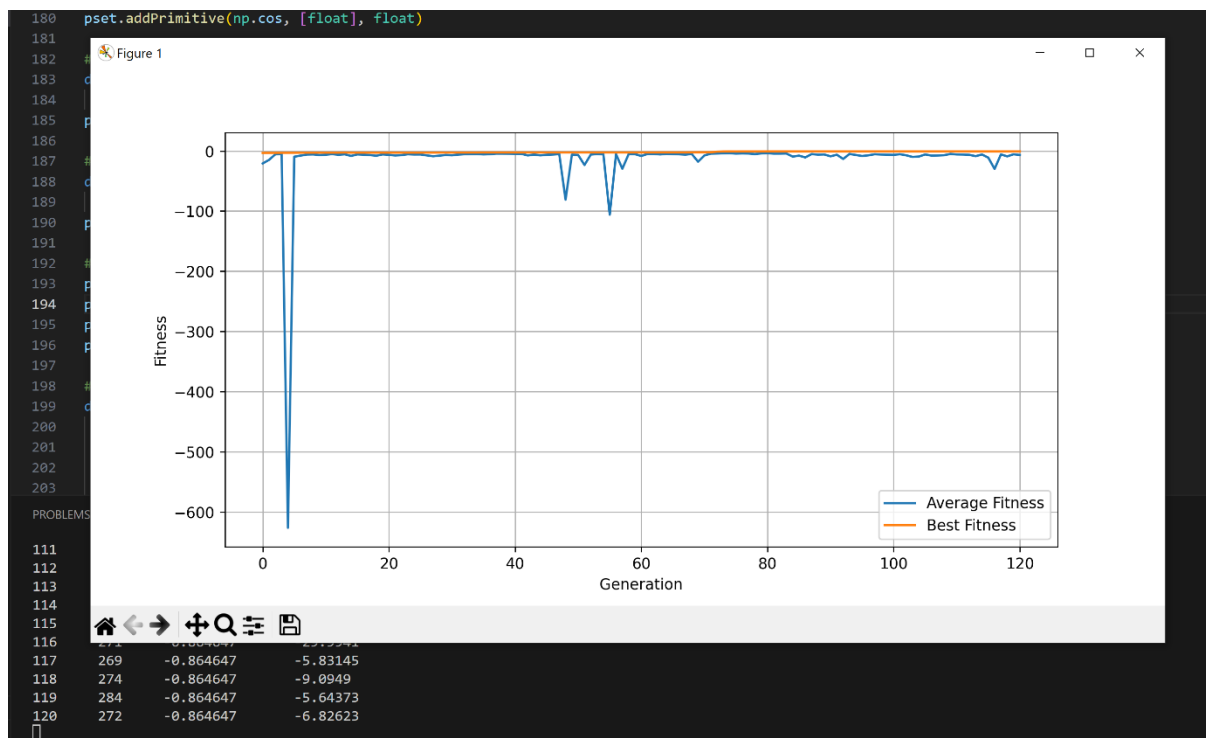
The fitness is calculated according to the following function:

$$\text{Fitness} = -(\text{MSE} + 0.001 * \text{size})$$

Test 1:

Max_depth	min_depth	Max_value	tournament size	number of generations	population size
2	4	6	5	120	300

Plot:



Output:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
114      266    -0.864647    -5.97048
115      266    -0.864647    -11.2096
116      271    -0.864647    -29.9941
117      269    -0.864647    -5.83145
118      274    -0.864647    -9.0949
119      284    -0.864647    -5.64373
120      272    -0.864647    -6.82623
best tree: add(if_then_else(is_positive(x1), sin(x2), 0.0), if_then_else(is_positive(x1), sin(x2), mul(cos(x1), x2)))
add(if_then_else(is_positive(x1), sin(x2), 0.0), if_then_else(is_positive(x1), sin(x2), mul(cos(x1), x2)))
Best fitness: -0.8646467176598377

```

Output explanation:

add(if_then_else(is_positive(x1), sin(x2), 0.0), if_then_else(is_positive(x1), sin(x2), mul(cos(x1), x2))), mul(cos(x1), x2))) =

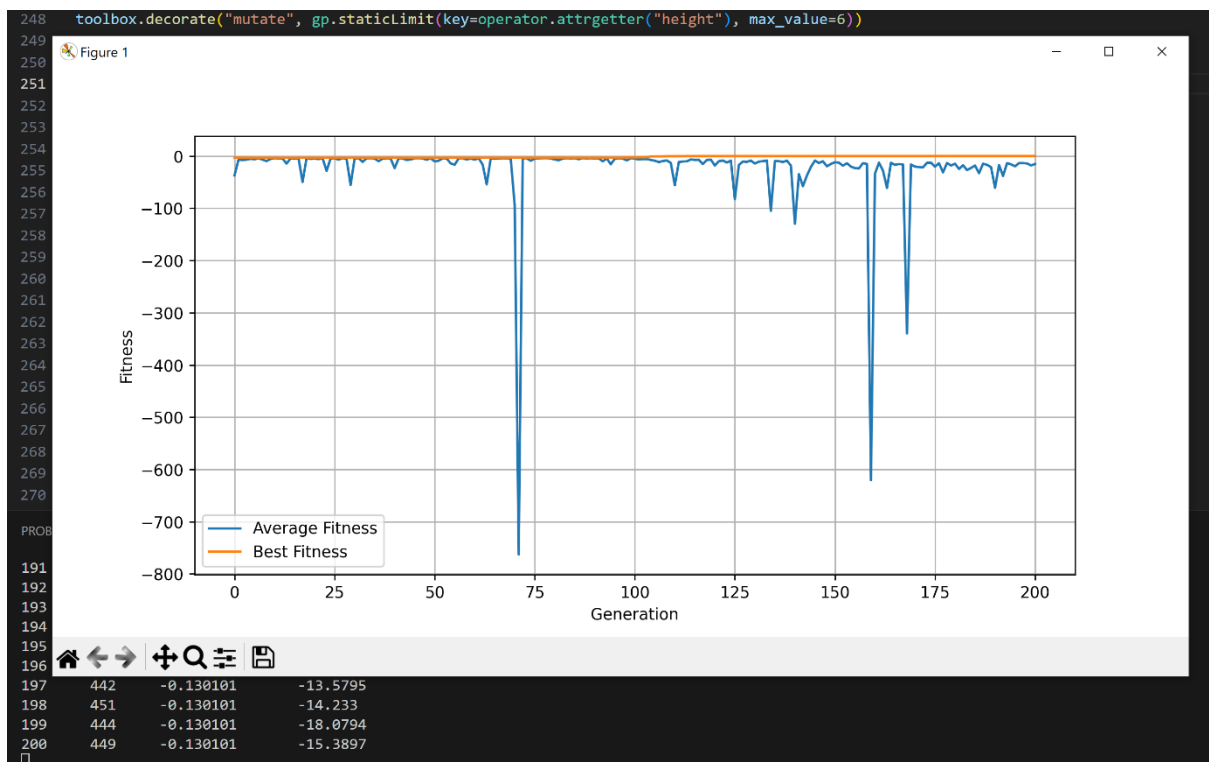
if x1> 0.0 then sin(x2) else 0.0 + if x1> 0.0 then sin(x2) else x2.cos(x1) = if x1> 0.0 then 2.sin(x2) else x2.cos(x1)

So the second part is correct.

Test 2:

Max_depth	min_depth	Max_value	tournament size	number of generations	population size
2	4	6	5	200	500

Plot:



Output:

```
def p2v12gen(x1):  
    PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
    194 443 -0.130101 -15.9933  
    195 445 -0.130101 -19.7068  
    196 449 -0.130101 -13.557  
    197 442 -0.130101 -13.5795  
    198 451 -0.130101 -14.233  
    199 444 -0.130101 -18.0794  
    200 449 -0.130101 -15.3897  
    best tree: mul(if_then_else(is_positive(x1), sin(x2), cos(x1)), if_then_else(is_positive(x1), x1, x2))  
    mul(if_then_else(is_positive(x1), sin(x2), cos(x1)), if_then_else(is_positive(x1), x1, x2))  
    Best fitness: -0.13010118082220523
```

Output explanation:

$$\text{mul}(\text{if_then_else}(\text{is_positive}(x1), \sin(x2), \cos(x1)), \text{if_then_else}(\text{is_positive}(x1), x1, x2)) =$$

$$(\text{if } x1 > 0.0 \text{ then } \sin(x2) \text{ else } \cos(x1)) * (\text{if } x1 > 0.0 \text{ then } x1 \text{ else } x2) = \text{if } x1 > 0.0 \text{ then}$$

$$x1.\sin(x2) \text{ else } x2.\cos(x1)$$

The entire expression is correct.